

Part I

Scientific Foundation

The very best professionals with expertise in low back injury prevention and rehabilitation have two things in common: an insight and wisdom developed from experience and a strong scientific foundation to enable evidence-based practice. Each problematic back is part of a whole person, who in turn is the sum of influences that range from those at the cellular level to those at the societal level. Some refer to this as the biopsychosocial approach. No two problematic backs (or patients) are the same, suggesting that optimal intervention would be different for each one. The best approach for optimal rehabilitation and prevention of those factors that exacerbate painful backs results from wise and scientifically based decisions. Part I of this book provides a foundation to enable you to make better decisions in your practice.

Chapter 1

Introduction to the Issues and Scientific Approach Unique to This Book

There is no shortage of manuals and books offering wisdom on low back health. Authors range from those with formal medical or rehabilitation training to laypeople who have found an approach to alleviate their own back troubles and become self-proclaimed low back health prophets. Their intentions are honorable, but their advice is rarely based on a sound scientific foundation. Too many of these authors offer inappropriate recommendations or even harmful suggestions. Years ago, as I began to develop scientific investigations into various aspects of low back problems, I would ask my graduate students to find the scientific foundation for many of the so-called commonsense recommendations I was hearing both in the clinic and in industrial settings. To my surprise they often reported that the literature yielded no, or very thin, evidence (note that I choose my students carefully and that they are very competent and reliable). Examples of such thinly supported “commonsense” recommendations include the following:

- Bend the knees to perform a sit-up.
- Bend the knees and keep the back straight to perform a lift.
- Take yoga and Pilates classes—they are good for the back.
- Reduce the load handled to reduce the risk of back troubles.
- Stretch the hamstrings if you have back pain or leg pain.

In fact, each of these recommendations may be appropriate in some situations but, as will be shown, not in all.

The famous economist John Kenneth Galbraith was well known for demonstrating that actions based on common wisdom, at least in economic terms, were often doomed to fail. He stated that common wisdom is generally neither common nor wise. Galbraith eloquently expressed exactly what I had experienced with “clinical wisdom” pertaining to the low back. Many attempts at preventing low back troubles and rehabilitating symptomatic ones have failed simply because they relied on ill-conceived clinical wisdom. This

history of failed attempts is particularly unfortunate because it has lent credence to the assertions of a number of increasingly well-known authorities that low back injury prevention and rehabilitation programs are a waste of resources. These authorities claim that the majority of low back problems are not organic at all—that, for example, most of these difficulties have materialized because workers are paid too much for injury compensation, have been subject to psychosocial influences, or crave sympathy. These dismissals of back injury are not justified. Back injury prevention and rehabilitation programs with strong scientific foundations, executions, and follow-up can be effective.

Having stated this, I must acknowledge that justifying improved practice based on scientific evidence is a dynamic process. With new evidence, the foundation will change. To account for such inevitable shifts, I have developed a balanced approach in these pages, reviewing the assets and liabilities and opposing views of an argument where appropriate. But I offer fair warning! As you read this book, be prepared to challenge current thoughts and rethink currently accepted practices of injury prevention and approaches to rehabilitation.

This chapter addresses some of the debatable issues regarding low back function, together with some opinions that rehabilitation professionals hold about patients, diagnosis, compensation, and disability. It also explores the circumstances that lead to back injury and discusses the need to apply this knowledge to improve low back injury prevention and rehabilitation.

Legislative Landscape: The Unfortunate Adverse Impact on Painful Backs

Although most legislation and legal activity involving back pain is enacted with good intentions, some has been counterproductive. A good example is the issue of spine range of motion (ROM). For years the American Medical Association (AMA) guidelines (1990) for quantifying the degree of back disability were based mostly on loss of spine ROM. Lawyers and compensation boards that needed numbers to define disability and award compensation latched onto spine ROM as an objective and easily measured factor. In the legal arena, therapy was considered successful when the ROM was restored or at least improved. This was changed a few years ago when range of motion was abandoned in favor of diagnosis-based estimates of disability (AMA, 2008). Now, charts link varying inability to perform functions of daily living, medical image findings, and clinically appropriate signs to specify impairment ratings. The contentious issues now include diagnosis of the specific condition, objectively quantifying pain, the existence of a candidate injury mechanism, whether exam findings are clinically relevant, and the ability of a medical image to show pathology. For example, a disc herniation often shrinks and grows depending on the prior loading history. How it appears on the day of the image acquisition may not represent the days in which pain is disabling.

These are scientifically supported and positive advances by the AMA. However, practice is much slower to change, and range of motion remains an emphasis in many clinics. After back injury, many people do not do well with an emphasis on enhancing spine mobility. In some cases, back problems are actually exacerbated by this approach. In fact, evidence shows that many back injuries improve with stabilizing approaches—motor control training, enhancement of muscle endurance, and training with the spine in a neutral position (Saal and Saal, 1989, may be considered the classic work). Our work showed that three-dimensional low back ROM has no correlation to functional test scores or even the ability to perform occupational work (Parks et al., 2003). Most recent work following soldiers in the Finnish military (Suni et al., 2013) showed that conscious training to reduce their lumbar range of motion during the conduct of duties reduced back pain over time. They used the principles in this book to reduce back pain incidence in a large

population. In the best practice, spine flexibility may not be emphasized until the very late stages of rehabilitation, if ever.

The idea of flexibility as the best measure of successful rehabilitation became entrenched for reasons of legal convenience rather than as a result of a positive impact on low back pain. It will take time for practice to fully recognize the importance of distinguishing between variables important for rating disability and those important for reducing pain and restoring ability. Another example illustrates the perverse impact of well-intended equity legislation. We all have equal rights under the law, but we are not physical equals. Although individual variance is present in every group, different populations within society demonstrate quite different capabilities. For example, the data of Jager and colleagues (1991), compiled from many studies, clearly showed that young men can tolerate more compressive load down their lumbar spines than can older men, and similarly, that men can tolerate about an additional third more load than can women when matched for age. Yet human rights legislation, which is designed to create fairness and equity by discouraging distinctions among groups, actually puts older females at greater risk than younger men. By not allowing a 64-year-old osteoporotic woman to be treated (and protected) differently from a 20-year-old, fit, 200 lb (90 kg) male in terms of tolerating spine load, the legislation presents a major barrier for intelligently implementing tolerance-based guidelines for protecting workers.

Deficiencies in Current Diagnostic Practices

It is currently popular for many authorities to suggest that back trouble is not a medical condition. They assert that physical loading has little to do with low back injury compensation claims; rather, they believe workers complain of back problems to benefit from overly generous compensation packages or to convince physicians that they are sick. According to this view, any biomechanically based injury prevention or rehabilitation program is useless. Variables within the psychosocial sphere dominate any biological or mechanical variable. If this is true, then this book is of no value—it should be about psychosocial intervention.

Those who contend that psychosocial factors dominate low back issues are well-published scientists and physicians. For example, Professor Richard Deyo (1998) summarized a common view: “Consider the following paradox. The American economy is increasingly postindustrial, with less heavy labor, more automation and more robotics, and medicine has consistently improved diagnostic imaging of the spine and developed new forms of surgical and nonsurgical therapy. But work disability caused by back pain has steadily risen.” This line of reasoning assumes that modern work (i.e., more repetitive, more sedentary) is healthier for the back than the predominantly physical labor of past generations. The evidence suggests, however, that the repetitive motions required by some specialized modern work, or the sedentary nature characterizing others, produces damaging biomechanical stressors. In fact, the variety of work performed by our great-grandparents may have been far healthier than our own. Deyo also seems to assume that nonsurgical therapy has been appropriately chosen for each person, whereas I suggest that inappropriate therapy prescriptions remain quite common.

Furthermore, although I agree that there is more reliance on medical imaging, I suspect that this reliance has resulted in a loss in the mechanically based diagnostic skills that are a crucial factor in accurate diagnosis. Interestingly, magnetic resonance imaging (MRI) has been documented to find—among other features—disc bulges, trophic facet joints, and “degenerative disc disease,” yet these have little relationship to whether the person has pain. The static images are commonly acquired when the person is in an unloaded, recumbent posture. Dynamic images (multiple X-ray, fluoroscopy, dynamic MRI) show much more movement pathology where

the dynamics can be directly related to pain. Discogenic pain, for example, has been shown to follow a natural history in which the zenith of pain is associated with the unstable phase, but the condition eventually ends with a very desiccated disc on a medical image and pain that has subsequently burned out. Then, abundant evidence shows, the history continues as changes in joint function influence the mechanical loading of the facet joints. Eventually, the facet joints become arthritic, shifting the pain source from disc to facet. Central sensitization further influences the process and links among pain, mechanical and functional factors, and patient corrections to wind down the heightened neural response. No wonder Savage, Whitehouse, and Roberts (1997) found little correlation between the image and the patient's symptoms.

Too often, the MRI impression or diagnosis is not correctly linked to the pain mechanism, and patients receive the incorrect treatment (Wassenaar et al., 2012), including surgery. Gibson, Martin, and Terry (1980) presented evidence regarding plain film radiographs; and McGill and Yingling (1999) and Zhao and colleagues (2005) discussed why such images are compromised in showing actual damage. Quite simply, bone displacement is needed for a radiograph to reveal damage. Yet we routinely observe quite substantial fractures that are held together by periosteum, which completely masks the damage.

Without question, the images are of great value for the surgeon who must “cut out the pain,” but only if the image finding corroborates the specific clinical symptoms of the patient. However, Chou and colleagues rightly pointed out that the images do not improve outcomes except in patients with severe neurological deficits (Chou, Deyo, and Jarvik, 2012). We have found that a thorough assessment reveals the cause of pain better than any image. The assessment is explained in chapter 9.

Professor Alf Nachemson (1992) wrote that “most case control studies of cross-sectional design that have addressed the mechanical and psychosocial factors influencing LBP (low back pain), including job satisfaction, have concluded that the latter play a more important role than the extensively studied mechanical factors.” Yet none of the several references cited to support this opinion made reasonable quantification of the physical job demands. Generally, these studies showed that psychosocial variables were related to low back troubles, but in the absence of measuring mechanical loading, they had no chance to evaluate a loading relationship.

Finally, Dr. Nortin Hadler (2001) has been rather outspoken, stating, for example, that “it is unclear whether there is any meaningful association between task content and disabling regional musculoskeletal disorders for a wide range of physical disorders” and that “on the other hand, nearly all multivariate cross sectional and longitudinal studies designed to probe for associations beyond the physical demand of tasks, detect associations with the psychosocial context of working.” Also, “the backache can be disabling nonetheless—not because of what is lifted, but whether or when it is to be lifted. Too often a salutary outcome proves elusive” (Hadler et al., 2007).

Recent evidence clearly shows that, although psychosocial factors can be important in modulating patient behavior, biomechanical components are important in the etiology of low back disorders and in their prevention. The position that biomechanics plays no role in back health and activity tolerance can be held only by those who have never performed physical labor and have not experienced firsthand the work methods that must be employed to avoid disabling injury. Although the scientific evidence is absolutely necessary, it will only confirm the obvious to those who have this experience. I find it perversely satisfying when physicians tell me that they are now, after missing work as a result of a nasty back episode related to physical work, able to relate to their patients. Perhaps experience with a variety of heavy work and with disabling pain should be required for some medics!

The implication of this literature is that the links between pain and biomechanical factors are variables over the natural history of back pain. Ignoring integrated natural history concepts obscures a mature interpretation of what the evidence and a thorough assessment really show: that the links among pain, biomechanics, and images are complex and patient specific, and that rigor is required to understand them.

It is, then, essential to investigate and understand the links among loading, tissue damage or irritation, psychosocial factors, and performance to provide clues for the design and implementation of better prevention and rehabilitation strategies for low back troubles. Founded on this rubric, chapters 8 and 9 strengthen the case for performing provocative testing to discover the cause of a patient’s pain and provide an algorithm to guide that approach. The following sections address several commonly held beliefs about back injury.

Are 85% of Back Troubles of Unknown Etiology?

Low back injury reports often mention the statistic that 85% of low back troubles are of unknown etiology. This has led to the popular belief that disabling back troubles are inevitable and just happen, a statement that defies the plethora of literature linking specific mechanical scenarios to specific tissue damage. Some have argued that this statement is simply the product of poor diagnosis—or of clinicians reaching the limit of their expertise (e.g., Finch, 1999). In fairness I must point out that diagnosis often depends on the profession of the diagnostician. Each group attempts to identify the primary dysfunction according to its particular type of treatment. For example, a physical therapist will diagnose based on manual therapy approaches, whereas a surgeon may find a diagnosis directed toward making surgical decisions more helpful. Some clinicians (surgeons, for example) seek a specific tissue as a pain candidate. From this perspective, nerve block procedures have shown conclusive pain source diagnoses in well over 50% of cases (e.g., Bogduk et al., 1996; Finch, 1999; Lord et al., 1996). This has prompted research into which tissues are innervated and are candidates as pain generators. Biomechanists often argue that this may be irrelevant because a spine with altered biomechanics has altered tissue stresses. Thus, a damaged tissue may cause overload on another tissue, causing pain whether the damaged tissue is innervated or not.

This is why other clinicians employ skilled provocative mechanical loading of specific tissues to reveal those that hurt, or at least to reveal loading patterns or motion patterns that cause pain. These types of functional diagnoses are helpful in designing therapy and in developing less painful motion patterns, but the process of functional diagnoses will be hindered by a poor understanding of spine biomechanics. Furthermore, those with a thorough understanding of the biomechanics of tissue damage can be guided to a general diagnosis by reconstructing the instigating mechanical scenario. An additional benefit of this approach is that, once the cause is understood, it can be removed or reduced. Unfortunately, many patients continue to have troubles simply because they continue to engage in the mechanical cause. Familiarity with spine mechanics will dispel this myth of undiagnosable back trouble and reduce the percentage of those with back troubles of no known cause.

Even with a tissue-based diagnosis, the practice of treating all patients who have a specific diagnosis with a singular therapy has not proven productive (Rose, 1989). For example, success rates with many cancer

therapies greatly improved with the combination of chemotherapy and radiotherapy. Optimal back rehabilitation requires the removal of the cause and the addition perhaps of stability, manual soft tissue therapy, or something else depending on the patient. Few patients fall into a complete fit for functional diagnosis in which a singular approach will yield optimal results. Both for interpretation of the literature and for clinical decision making, it would appear prudent to question the diagnostic criteria needed before assigning a diagnosis.

Limitations in tissue-based diagnosis should not be used to suggest that determining the cause of back troubles is irrelevant or that the manual or medical treatment in some cases is fruitless, leaving psychosocial approaches to prevail by default. As presented later in this book, identifying the motions, postures, and loads that cause pain, or relieve pain, results in a precise diagnosis. The levels of pain triggers are determined. Understanding spine function enables links between pain patterns and causation to be established. This diagnostic approach is productive for guiding prevention and rehabilitation approaches. But what is meant by this approach? Read on!

Diagnosis by Hypothesis Testing

The opinion that precise diagnosis for back pain is not possible is now old-fashioned. Geoff Maitland (1987), the Australian physiotherapist, years ago promoted the concept of examining the patient and forming a working hypothesis. The hypothesis was used to guide treatment and project the prognosis. The hypothesis was then tested and refined as rehabilitation progressed. Our approach, which incorporates a strong biomechanical foundation and blends expertise from various biomedical and psychosocial disciplines, is strongly aligned with Maitland's proposition. An initial impression is formed from the first meeting of patients in the waiting room—from observing their sitting posture, how they rise from the chair, their initial gait pattern, and so on. Then a history is taken to look for possible candidate injury mechanisms as well as perceived pain exacerbators and relievers. Observation continues during some basic motion patterns as the evaluation process proceeds, delving further into the mechanics and nature of the symptoms. Then provocative tests are performed to either strengthen or weaken the hypothesis. Motion and motor patterns that are tolerated are

identified. All information is used to formulate the plan for corrective exercise and the starting dosage of tolerable therapeutic exercise. The progression concludes with functional screens and tests that are chosen based on information obtained in the preceding process. In this way a functional diagnosis is ensured that is sufficient for considering exercise choice and rates of subsequent progression. In some cases the tissue causing the pain is clearly identified. In others the cause is precisely identified in terms of specific motions, postures, and loads together with symptoms of abnormal motion and motor patterns. The course of prevention and rehabilitation now has some quantitative guiding parameters. The patient and clinician are able to eliminate the variables that cause pain and build progressions that avoid pain.

Are Most Chronic Back Complaints Rooted in Psychological Factors?

Although there is no doubt that many chronic back cases have psychological overlays, the significance of psychology for back problems is often greatly exaggerated. Dr. Ellen Thompson (1997) coined the phrase bankrupt expertise when referring to spine docs who are unable to guide improvement in their patients and default to blaming the patients and their psychoses. These physicians either dismiss mechanical causation or assume that mechanical causation has been adequately addressed. These are the ones who will state that the pain is in the patient's head or that the patient is noncompliant with the ill-prescribed therapy.

At our university clinic I see patients who have been referred by physicians for consult. These are either elite performers or those with very difficult chronic bad backs who have failed with all other approaches. In spite of the fact that these people have received very thorough attention, I am continually heartbroken to hear about the minimal notice paid to ongoing back stressors and about the exercises that these “basket case backs” have been prescribed that have only exacerbated their conditions. The day before I wrote this section, I saw a classic example.

A woman had suffered for 5 years on disability and had seen no fewer than 12 specialists from a variety of disciplines. Although several had acknowledged that she had physical concerns, her troubles were largely

attributed to mental depression. She consistently reported being unable to tolerate specific activities while being able to tolerate others. Some provocative testing confirmed her report and uncovered a previously undiagnosed arthritic hip. For years she had been faithfully following the instructions of her health care providers to perform pelvic tilts, knees-to-chest stretches first thing in the morning, and sit-ups; to take her large dog for walks; and so on. All of these ill-chosen suggestions had prevented her posterior disc (with sciatica)-based troubles from improving. As we will see later, these types of troubles typically do not recover with flexion-based approaches—particularly first thing in the morning. Moreover, the lead-imposed torsional loads that she experienced every time she walked her dog exceeded her tolerance. Although she reported vacuuming as a major exacerbator of her troubles, her health care providers had never shown her how to vacuum her home in a way to spare her back. I suggested that removing these daily activities and replacing the flexion stretches with neutral spine position awareness training and isometric torso challenges would likely start a slow, progressive recovery pattern. I believed that her psychological concerns would probably disappear with her back symptoms if she fell into the typical pattern. This patient, with this typical story, has a reasonably good chance to enjoy life once again. (Note: This patient was back to work and off her antidepressant medication at the time of proofing this manuscript.)

None of the experts this woman had seen—including physical therapists, chiropractors, psychologists, physiatrists, neurologists, and orthopedic surgeons—addressed mechanical concerns. This is not to condemn these professions, but rather to suggest that sharing experiences and approaches will help us to be more successful in helping people with bad backs. Perhaps these professionals were unaware of the principles of spine function, the types of loads that are imposed on the spine tissues during certain activities, and how these activities and spine postures can be changed to greatly reduce the loads—in other words, the biomechanical components. The links between the physical and the psychological variables are becoming better understood. Virtually all of these are well addressed with mechanical approaches. For example, reduction in pain-catastrophizing behavior reduces just as effectively with mechanical approaches as it does with the usual cognitive behavioral therapy approaches (Smeets et al., 2006).

This book is an attempt to heighten the awareness and potential of this

mechanical approach. Although it sounds very harsh, I have found relatively few experts who appear willing to adequately address the causes of back troubles while working to find the most appropriate therapy. My years of laboratory-based work, combined with collaboration with my clinical colleagues, have provided me with unique insight. As a result, I am not so quick to blame the chronic patient.

Does Pain Cause Activity Intolerance?

Evidence that mechanical tissue overload causes damage is conclusive. But does the damage cause pain, and does the chronic pain cause work intolerance? Although this seems intuitive, the research needed to confirm these links is difficult to perform. Several, but limited numbers of, studies have documented the mechanical or chemical stimulation of tissues to reproduce clinical pain patterns. (The absence of definitive, large-scale studies is due to the ethical issues of performing invasive procedures and probably not to lack of scientific merit in such studies.) For example, the pioneering work of Hirsch and colleagues (1963-64) documented pain from the injection of hypertonic saline into specific spine tissues thought to be candidates for damage. Subsequent to this work, several other studies suggested the link between mechanical stimulation and pain—for example, the work of Hsu and colleagues (1988) documenting pain in damaged discs. Ortiz and Bordia (2011) were able to document the links among loading, damage, and resulting pain in those who develop osteoporotic vertebral compression fractures. A preexisting degenerated disc disrupts the normal axial transfer of load, thus acting as the injury vector resulting in compromised mechanics and adjacent compression fractures, and pain.

There is irrefutable evidence that vertebral disc end-plate fractures are very common and result only from mechanical overload (Brinckmann, Biggemann, and Hilweg, 1989; Gunning, Callaghan, and McGill, 2001). That these fractures are also found in necropsy specimens that were subjected to whiplash (Taylor, Twomey, and Corker, 1990) also strengthens another facet of this relationship. Hsu performed discograms (injections of radio contrast) into 692 discs, of which 14 demonstrated leakage into the vertebral body, confirming an end-plate fracture. Four of these discs (28%) produced severe pain, nine (64%) produced fully concordant pain, and one produced mildly

discordant pain. In contrast, only 11% of the remaining 678 discs with no end-plate disruption produced severe pain, 31% produced concordant pain, 17% produced mild pain, and 41% produced no pain. This evidence provides strong support for the notion that loading causes damage and damage causes pain.

Even though pain can limit function and activity in other areas of the body, some still suggest that these are not linked when a bad back is at issue. Teasell (1997) provided an interesting perspective when he argued that in sports medicine, as opposed to occupational medicine, it is well accepted that some injuries require months of therapy or can even cause retirement from the activity. He noted that athletes receiving specialized sports medicine care are an interesting group to consider because many are highly motivated, are in top physical condition, are well paid, have access to good medical care, and are fully compensated even while injured. Their injuries and pain can cause absence from play for substantial amounts of time and can even end their lucrative careers. Teasell reminded us that not all long-term chronic pain is an entirely psychosocial concern, as implied by some clinicians. These clinicians' dismissal of the usefulness of physical approaches simply because they have not been successful in reducing long-term troubles is a disservice to the patient.

Inadequacies in Current Care and Prevention

Many back patients can testify that the care they have received for their troubles is not satisfactory. Conversations with family doctors have left me with the general impression that they do not know what to do with the back pain case except to provide analgesics and perhaps a referral to a surgeon or therapist. They have neither the time nor the expertise to provide specific help matched to the patient. Suggesting that they engage in a nonspecific activity such as yoga may be just as helpful as harmful—the net gain is zero. Carey and colleagues (2009) documented practice patterns in 5,357 households in North Carolina. Of the 732 adults with chronic LBP, 60% had used narcotics in the previous month and over one-third had advanced imaging in the previous year, yet only 3% had engaged in formal spine rehabilitation. The discrepancy between the overemphasis on expensive treatments and the underutilization of pain-reducing movement and exercise is clear.

What are some of the factors that contribute to the inadequacy of patient experience? Certainly one is the fact that the epidemiological evidence on which many professionals base their treatment recommendations can be quite confusing. Following are some examples of the issues that cause confusion.

- Plethora of studies on “backache.” Nonspecific backache is nearly impossible to quantify and, even if it could be quantified, offers no guidance for intervention. As such, any study of treatment interventions on nonspecific backache is of little use. Some people suffer from discogenic problems, for example, and will respond quite differently from those with ligamentous damage or facet-based problems. Efficacy studies that do not subclassify bad backs end up with nonspecific average responses. This has led to the belief that nothing works—or that everything does, but to a limited degree. More studies on nonspecific backache treatment will not be helpful, nor will the large epidemiological reviews of these studies offer real insight. In contrast, patients with treatments matched to their conditions experience greater short- and long-term reductions in disability than those receiving unmatched treatments (Brennan et al., 2006; Fritz, Cleland, and Childs, 2007).

- U-shaped function of loading and resulting injury risk. Like many health-related phenomena, the relationship of low back tissue loading to injury risk appears to form a U-shaped function—not a monotonically rising line. For example, virtually every nutrient will cause poisoning with excessive dosage levels, but health suffers in their absence; thus, there is a moderate optimum. In the case of low back loading, evidence suggests that two regions in the U-shaped relationship are problematic—too much and too little. Porter (1987) suggested that heavy work is good for the back—but how does one define heavy? Porter was probably referring to work of sufficient challenge and variability to reach the bottom of the U and hence lower symptoms. From a biological perspective, sufficient loading is necessary to cause strengthening and toughening of tissues, but excessive levels will result in weakening. In other words, it is not a matter of doing exercise or not, or engaging in activity or not. Heneweer, Vanhees, and Picavet (2009) expressed the need to match the appropriate activity and its level for seeking the optimal point on the U-shaped relation for back pain.
- Relationship of intensity, duration of loading, and rest periods. As Ferguson and Marras (1997) pointed out, some studies suggest that a certain type of loading is not related to pain, injury, or disability, whereas others suggest it is, depending on how the exposure was measured and where the moderate optimum for tissue health resides for the experimental population. The subjectivity of such studies is further underscored when we consider the question of whether there is a clinical difference between tissue irritation and tissue damage. Loading experiments on human and animal tissues to produce damage reveal the ultimate tolerable load beyond which injuries cause biomechanical changes, pain, and gross failure to structures. In real life, any of us could irritate tissues to produce tremendous pain at loading levels well below the cadaver determined tolerance by repeated and prolonged loading. In fact, evidence presented by Videman and colleagues (1995) and Niemelainen and colleagues (2008) suggests that the progressive development of conditions such as spinal stenosis results from years of specific subfailure activity. The fundamental question is Could such conditions be avoided by evidence-based prevention strategies that include optimal loading, rest periods, and controlling the duration of exposure?

III-Advised Rehabilitation Recommendations

These failures to appropriately frame research and its results have resulted in many oversimplifications about low back treatment, which have in turn led to some inadequate treatment practices and recommendations. A few of the most common recommendations for back health are discussed here.

- Strengthen muscles in the torso to protect the back. Despite the clinical emphasis on increasing back muscle strength, several studies have shown that muscle strength cannot predict who will have back troubles (Biering-Sorenson, 1984). On the other hand, Luoto and colleagues (1995) and Stroyer and Jensen (2008) have shown that muscular endurance (as opposed to strength) is protective. Why, then, do many therapeutic programs continue to emphasize strength and neglect endurance? Perhaps it is a holdover influence from the athletic world, in which the goal of training is to enhance performance. Perhaps it is an influence from the pervasive use of bodybuilding approaches in rehabilitation. As will be shown, optimal exercise therapy occurs when the emphasis shifts away from the enhancement of performance and toward the establishment of improved health. In many cases the two are mutually exclusive!
- Bend the knees when performing sit-ups. Clinicians widely recommend bending the knees during a sit-up, but on what evidence? A frustrating literature search suggests that this perception may be the result of so-called clinical wisdom. Interestingly, Axler and McGill (1997) demonstrated that there is little advantage to one knee position over the other, and in fact the issue is probably moot because there are far better ways to challenge the abdominal musculature and impose lower lumbar spine loads (traditional sit-ups cause spine loading conditions that greatly elevate the risk of injury). This issue is one of many that will be challenged in this book.
- Performing sit-ups will increase back health. Is this a true statement or an artifact of experimental methodology? Despite what many would like to believe, there is only mild literature support for the belief that people who are fit have less back trouble (although positive evidence is increasing—for example, Stevenson et al., 2001). Interestingly, many of the studies attempting to evaluate the role of increased fitness in back

health actually included exercises that have been known to cause back troubles in many people. For example, many have attempted to enhance abdominal strength with sit-ups. After examining the lumbar compression that results from performing sit-ups with full flexion of the lumbar spine, together with excessive disc annulus stresses, it is clear that enough sit-ups will cause damage in most people. Each sit-up produces low back compression levels close to the U.S. National Institute for Occupational Safety and Health (NIOSH) action limit, and repeatedly compressing the spine to levels higher than the NIOSH action limit has been shown to increase the risk of back disorders (Axler and McGill, 1997). Thus, reaching a conclusion about the role of fitness from the published literature has been obscured by ill-chosen exercises. Increased fitness does have support, but the way fitness is increased appears to be critical.

- To avoid back injury when lifting, bend the knees, not the back. Probably the most common advice given by the clinician to the patient who must lift is to bend the knees and keep the back straight. In addition, this forms the foundation for virtually every set of ergonomic guidelines provided to reduce the risk of work-related injury. Very few jobs can be performed this way. Further, despite the research that has compared stooping and squatting styles of lifting, no conclusion as to which is better has been reached. The issue of whether to stoop or squat during a lift depends on the dimensions and properties of the load, the characteristics of the lifter, the number of times the lift is to be repeated, and so forth, and there may in fact be safer techniques altogether. Much more justifiable guidelines are provided later in this text.
- Tight hamstrings and unequal leg length lead to back troubles. It would seem intuitive that shortened, or tight, hamstrings would apply deleterious torque to the pelvis and lead to back troubles. A similar argument could be mounted for unequal leg length, which would tilt the pelvis and impose bending stresses on the lumbar spine. This line of reasoning appears to have driven popular clinical practice. Interestingly, there is little support for these notions. A longitudinal study of young men throughout their military service did not reveal any link between current back pain and hip flexion restrictions (interpreted as hamstring tightness) (Hellsing, 1988). Neither could pain be predicted in this study or in the well-conducted studies of Biering-Sorenson (1984) and Van

Nieuwenhuyse and colleagues (2009) on the Belgian Low Back Cohort. Developing pain while standing was not linked to hamstring extensibility (Raftry and Marshall, 2012). An interesting study by Ashmen, Swanik, and Lephart (1996) suggested that although reduced hip flexion may not be associated with back pain, asymmetry between sides may be. Interestingly, many high-performance athletes who run and jump have tight hamstrings that they use as springs. Note that hip joint restriction is a different issue; evidence from several perspectives links hip pathology with back symptoms. For example, Cibulka and colleagues 1998 noted the association between internal/external rotation and sacroiliac dysfunction and back pain. Our work has surveyed restricted hip rotation in workers with demanding jobs and chronic recurrent back pain (McGill et al., 2003). Unequal leg length has been shown to have a link with back pain in only the most extreme of length discrepancies; even those with a 2 in. (5 cm) difference rarely develop chronic pain (Grundy and Roberts, 1984). Further, a significant link between leg length inequality and lumbar scoliosis does not seem to exist, at least for inequalities of 0.4 in. (1 cm) or less (Hoikka, Ylikoski, and Tallroth, 1989). All of this suggests caution when assessing patients and attributing symptoms. Perform the provocative tests and discover whether these postural variables are true exacerbators in the individual and thus are justifiable targets for therapy.

- A single exercise or back stability program is adequate for all cases. It is currently popular to promote the training of single muscles to enhance spine stability. Although the original research was motivated by the intention to reeducate perturbed motor patterns that were documented to be the result of injury, others have misinterpreted the data and are promoting exercises to train muscles they believe are the most important stabilizers of the spine. Unfortunately, they did not quantify stability. The process of quantifying the contribution of the anatomical components to stability reveals that virtually all muscles can be important, but their importance continually changes with the demands of the activity and task. It is true that damage to any of the spinal tissues from mechanical overload results in unstable joint behavior. Because of biomechanical changes to the joint, however, the perturbed tissue is rarely linked to the symptomatology in a simple way. More likely, other tissues become involved, and which ones are involved will result in

different accompanying motor disturbances. This variety in possible etiologies means that a single, simple rehabilitation approach often will not work. Is mobility to be restored at the expense of normal joint stability? Or is stability to be established first, with enhanced mobility as a secondary delayed rehabilitation goal? Or is the clinical picture complex—for example, when spine stability is needed but tonic psoas activity causing chronic hip flexure necessitates hip mobilization? This example, one of many that could have been chosen, illustrates the challenge of ensuring sufficient stability for the spinal tissues. No simple, or single, approach will produce the best results in all cases. The description and data presented in this book will help guide the formulation of exercises that ensure spine stability.

Can Back Rehabilitation Be Completed in 6 to 12 Weeks?

Some have suggested that damaged tissues should heal within 6 to 12 weeks. In fact, many have used this argument to support the notion that work intolerance exceeding this period has no pathoanatomical basis (e.g., Fordyce, 1995) but stems from psychosocial issues. Further, some have suggested that patient recovery would be better served by redirecting rehabilitative efforts away from physical approaches. This position can be refuted by data and indicates a misunderstanding of the complexities of spine pathomechanics. The concept that tissues heal within 6 to 12 weeks appears to be originally based on animal studies (reviewed in Spitzer, 1993), the majority of which were of rodent muscle. However, not all human patients get better so quickly (Mendelson, 1982), and the follow-up studies from some defined disorders such as whiplash are compelling in the support of lingering tissue disruption (e.g., Radanov et al., 1994; Curatolo et al., 2011). A systematic review by Itz and colleagues (2012) challenged the claim that most first-time back pain episodes resolve in the first 6 weeks. They showed that spontaneous recovery is not true because 65% of these people reported back pain after 1 year.

Evidence is presented in later chapters of both mechanical and neurological changes that linger for years subsequent to injury. This includes loss of various motor control parameters together with documented

asymmetrical muscle atrophy and other disorders. This suggests that postinjury changes are not a simple matter of gross damage healing. Following are only a few of the types of damage that can be long-term indeed:

- Specific tissues such as ligaments, for example, have been shown to take years to recover from relatively minor insult (Woo, Gomez, and Akeson, 1985).
- The intervertebral motion units form a complex mechanism involving an intricate interplay among the parts such that damage to one part changes the biomechanics and loading on another part. From the perspective of pathomechanics, many reports have documented the cascade of biomechanical change associated with initial disc damage (Galbusera et al., 2011) and subsequent joint instability and secondary arthritis, which may take years to progress (e.g., Brinckmann, 1985; Kirkaldy-Willis, 1998).
- Videman and colleagues (1995) and Niemelainen and colleagues (2008) documented that vertebral osteophytes were most highly associated with end-plate irregularities and disc bulging. Osteophytes are generally accepted to be secondary to disc and end-plate trauma but take years to develop.

Thus, to suggest that back troubles are not mechanically based if they linger longer than a few months only demonstrates limited expertise.

Another question is Can these back troubles linger for a lifetime? In this connection, it is interesting that elderly people appear to complain about bad backs less than younger people do. Valkenburg and Haanen (1982) showed that back troubles are more frequent during the younger years. Weber (1983) provided further insight by reporting on patients 10 years after disc herniations (some of them had had surgery, whereas others had not) who were engaged in strenuous daily activity—yet all were still receiving total disability benefits! It would appear that the cascade of changes resulting from some forms of tissue damage can take years, but generally not longer than 10 years. Although the bad news is that the affected joints stiffen during the cascade of change, the good news is that eventually the pain is gone.

To summarize, the expectation that damaged low back tissues should heal within a matter of weeks has no foundation. In fact, longer-term troubles do

have a substantial biomechanical or pathoanatomical basis. On the other hand, troublesome backs are generally not a life sentence.

Should the Primary Goal of Rehabilitation Be Restoring Range of Motion?

Research has shown that an increased ROM in the spine can increase the risk of back troubles (e.g., Battie et al., 1990; Biering-Sorenson, 1984; Burton, Tillotson, and Troup, 1989). Why, then, does increasing ROM remain a rehabilitation objective? The first reason, discussed earlier, is the need to quantify reduced disability as defined by the AMA. Second, there is a holdover philosophy from the athletic world that increased ROM enhances performance. This may be true for some activities, but it is untrue for others. As will be shown, this philosophy may work for other joints, but it generally does not work for the back (note that loss of hip mobility contributes to eventual low back disorders; the specific loss has a specific effect on the spine [Van Dillen et al., 2008]). In fact, successful rehabilitation for the back is generally retarded when athletic principles are followed.

When Is Surgery Justifiable?

In practice I see the people who are worse off because of surgery; surgeons refer to them as failed backs (note: I do not see the surgical successes). I strongly suspect that many had the surgery too soon. They thought they had exhausted conservative approaches, but a review of their files revealed that they had received inappropriate treatment. The therapeutic exercises prescribed to them were not matched to their specific pain mechanisms and caused more pain. They were not shown the specific causes of their pain, so they failed in preventing pain. Or their surgeon's declaration that they "failed conservative care" simply meant that a specific amount of time had passed.

There are assets and liabilities to surgery and the particular approach. Whereas fusion has been shown to increase adjacent segment level facet arthritis, disc replacement has been shown to spare them, on average (Siepe et al., 2010). However, it is noted that facet degeneration at the level of the replacement is increased (Hellum et al., 2012), suspected to be due to the

natural center of joint rotation not replicated by the prosthesis, which results in stress concentrations on the facet. Fusion is intended to stop motion and pain, whereas replacement is intended to restore motion and stop pain. Justifiable guidance for patient selection remains lacking. Note that there were still failures resulting from both approaches.

Vertebroplasty is a technique of injecting cement into a fractured osteoporotic vertebra with the intention of relieving pain and restoring load-bearing ability. Buchbinder and colleagues (2009) studied over 70 patients and questioned the efficacy of this approach. Microdiscectomy techniques are intended to reduce tissue disruption for disc herniation and nerve root decompression. Arts and colleagues (2009) documented a well-conducted trial comparing tubular discectomy (operating through a scope tube) with conventional discectomy. Once again, the small changes favored conventional microdiscectomy techniques for pain reduction, although the outcome would be heavily influenced by surgeon familiarity and skill, and patient selection. Obviously, given that the tubular approach is less invasive, the risk of inadvertently disrupting the dura and other critical tissues is greater because of the reduced vision of the area. Conversely, the greater the surgical exposure with the conventional approach, the greater the gross tissue damage; however, sensitive tissues are better seen and thus protected. Abdu and colleagues (2009) studied pain from spondylolisthesis by following three methods of hardware-based fusion. They concluded that the patients were better off after the surgery, but there was no difference among approaches.

Follow-up of postsurgical people compared with nonsurgical people can be sobering. Nguyen and colleagues (2011) followed up a historical group of compensated back-injured workers over the 3-year period from 1999 to 2001. A total of 725 had had lumbar fusion, compared with 725 nonsurgical control back-injured workers. After 2 years, 67% of nonsurgical people had returned to work compared with 26% of those who had had surgery. Reoperations were conducted on 27% of the surgical cases, and 36% reported complications and higher long-term opioid use. Permanent disability occurred in 11% of postsurgical patients and 2% of nonsurgical patients. The notion that surgery will eliminate back pain does not have statistical support. People have to manage their postsurgical backs just as much as they would their nonsurgical backs.

When interpreting any of these studies, the key is understanding the variability of response, not only the average effect of the surgical technique.

With every technique, some patients got better; and others, worse. Be aware of the selection of patients, which is usually quite defined for each study. Understanding the variables that explain this divergent response will lead to better outcomes in the future. Once again, patient selection appears to be of paramount importance.

Following is a set of rules that I compiled that I give to patients considering surgery:

- First, always try the virtual surgery game and consider surgery only when it fails. Some patients believe that daily workouts will cure their back pain and that failing to do so will cause weight gain and more pain. Usually, it is the exercises and movement that are keeping them patients! In reality, they need to stop irritating and sensitizing the pained tissues. The virtual surgery game is this: Pretend you had surgery today. Tomorrow is the first day of recovery and is characterized by gentle movements and activities, but generally it is a day of forced rest. Subsequent days follow a typical postsurgical progression in restricted activity. The positive response in symptoms, on average, is impressive.
- Consider surgery when neurological issues are substantial, such as loss of bowel and bladder control. This does not include radiating symptoms such as sciatica. Radiating pain symptoms, peripheral numbness, muscle atrophy, and so on, are all signs of trapped or compromised spine nerve roots. We have been successful in each of these conditions using nerve mobilization approaches. Always try these first. Caution: These are special techniques that require expertise. Performed improperly or too aggressively, they will increase symptoms. Approaches to reduce the cause of the nerve compression or irritation are also required in stubborn cases (see the discussion of nerve flossing techniques in chapter 10).
- Consider surgery in cases of trauma. In such cases, the structure is unstable and needs stabilizing.
- Consider surgery only when pain has been unrelenting and severe for a substantial period of time. Patients who have severe pain for 3 weeks and then have surgery have been some of the most disabled postsurgical cases I have seen.
- Select the surgeon. Everyone likes to state that they had the best surgeon. I have observed sloppy and uncaring work, yet these surgeons continue to practice. I have found that asking the nurses and physical

therapists at the hospital which surgeon has the best results is a wise approach. Simply because a surgeon is the head of the department or may be speaking at a medical conference does not indicate the skill in his hands.

- Discuss the pain with the surgeon. What is the pain generator, and can she cut it out? If several tissues are involved, the chances for success go down. If there is damage at several spine levels, the chance for success drops substantially.
- Clarify what the success rate is. The word success is loaded. In some medical reports this means the patient did not die. In others it means that the patient did well for a brief period following the operation. You are most interested in the long-term success rate compared to any other option, as well as the risks and benefits.
- Beware of new treatments. I have observed so many new devices that have been put into people who end up with poor long-term results. Discs have been stiffened with papain injections, heated with a catheter to stiffen them, and cored and screwed with titanium cages, to name a few. None of these lived up to initial claims (read an insightful review of surgical interventions by Carragee and colleagues, 2009).
- Beware of disc replacement, an approach that I am very skeptical of. I have not seen a successful case yet—that is, full resumption of former activities without pain and the ability to bear substantial load. Most spine joint pain results from motion and loading. Traditionally, the surgeons fuse the joint in a variety of ways. But an artificial disc does the opposite. Its goal is to restore motion to the joint. But herein lies the flaw in logic. The disc is just one of three joints at each spine level. The artificial disc creates an axis of rotation that does not mimic the natural axes of the natural disc. This places more stress on the other two facet joints, so that, over time, they become arthritic and intolerant to motion. Long-term spine health is nearly impossible to achieve this way. The process of patient selection for disc replacement also puzzles me. I have reviewed images of spines with replacements at one level; however, there was clear evidence of substantial instability at multiple levels with large traction spurs and the like. What sort of selection criteria led to a surgical opinion that replacement at a single level would take the pain away? My advice now: Ask the surgeon to set up a conversation with a couple of former patients so that you can be assured of their satisfaction.

- Always exhaust the conservative options. You may believe that because you tried physical therapy, or any other approach, and it failed, only a surgical option remains. It may be that the therapy was not appropriate for you.
- Beware of institutes that offer to view medical images and, with no other information, advise patients on surgery. Pictures are not linked to pain; a thorough clinical assessment is absolutely essential.

I must conclude by noting that there are excellent surgeons. I am familiar with one who insists that any potential patient sign a contract with him. The first condition is that the patient agrees to engage in a structured preoperative program of a specified duration per day and per week, incorporating principles similar to those outlined in this book. The second condition is that if the outcome from this approach is not satisfactory and surgery occurs, the patient also agrees to postsurgical rehabilitation of a certain duration each day to enhance spine-sparing biomechanical principles and pain-free function. That is outstanding leadership!

Several essays in the medical literature assert that back pain is not a medical problem but a social problem. These authors claim that patients are catastrophizing their pain. I disagree. I see the problem as being with the system and the practitioners. Medical practitioners will never be motivated to enhance their understanding of mechanical causes, appropriate diagnoses, and treatments when surgeons who operate are paid 100 times what they would be paid to consult with the patient about what causes pain. Patients often get 3 or 4 minutes for a consult and similar attention for preoperation consults. The practice of using medical images as indicators of pain has replaced a thorough examination that would reveal the pain-causing mechanisms. Only a substantial understanding of biomechanical function and the variables that influence a patient's presentation will result in the best treatment approach.

A Better Alternative to Dealing With Painful Backs

Painful backs are the result of a variety of causes; this book proposes approaches to identify the cause in the individual. Understanding the specific cause, or exacerbator, in each person directs efforts to remove the cause, and ensures that the cause is not replicated in the therapy. This approach works

(Ikeda and McGill, 2012). Tissues in the back become irritated with repeated loading. Consider accidentally stubbing a toe or biting the lip repeatedly—eventually the slightest touch causes pain. This is symptom magnification because the tissues are hypersensitized—not because of psychosocial modulators. Reduction of hypersensitivity in the toe or lip only occurs following a substantial amount of time after the accidental stub or bite has stopped. Tissues in the back are continually “hit” because of aberrant motion or motor patterns. For example, people with flexion bending intolerance of the spine may replicate this every time they rise from a chair. Correcting this movement fault, metaphorically taking the hits away, results in less sensitized tissues, an increased repertoire of pain-free tasks, and a return of motion. Motion returns once the pain goes away. Resist attempting to restore function with a mobilizing approach too soon. This often retards progress.

Mechanical Loading and the Process of Injury: A Low Back Tissue Injury Primer

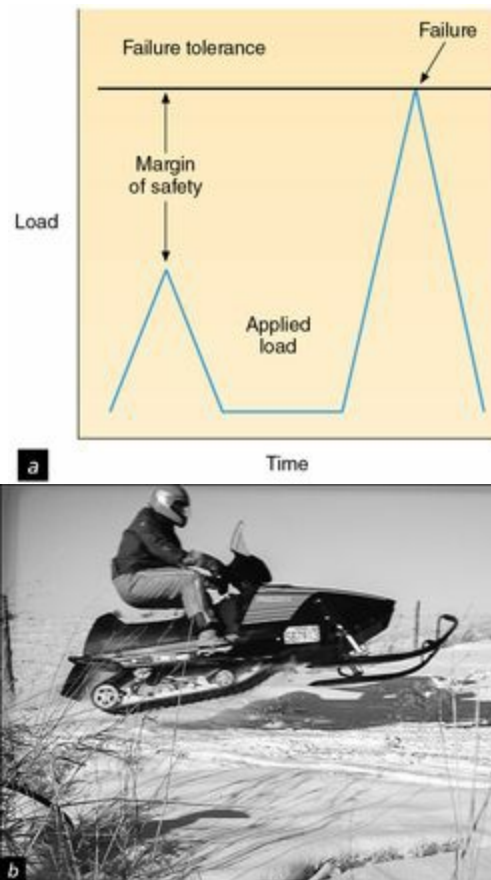
Any clinician completing a worker or patient compensation form is required to identify the event that caused the injury. Very few back injuries, however, result from a single event. This section documents the more common cumulative trauma pathways leading to the culminating event of a back injury. Because the culminating event is falsely presumed to be the cause, prevention efforts are focused on that event. This misdirection of efforts fails to deal with the real cause of the cumulative trauma.

Although a generic scenario for injury is presented here, chapter 4 offers a more in-depth discussion of injury from repeated and prolonged loading to specific tissue. The purpose of this section is to promote the consideration of the many factors that modulate the risk of tissue failure and to encourage probing to generate appropriate hypotheses about injury etiology.

Injury, or failure of a tissue, occurs when the applied load exceeds the failure tolerance (or strength of the tissue). For the purposes of this discussion, injury is defined as the full continuum from the most minor of tissue irritation (but microtrauma nonetheless) to the grossest of tissue failure, such as vertebral fracture or ligament avulsion. We will proceed on the premise that such damage generates pain.

Obviously, a load that exceeds the failure tolerance of the tissue, applied once, produces injury (see [figure 1.1b](#), in which a Canadian snowmobiler airborne and about to experience an axial impact with the spine fully flexed is at risk of posterior disc herniation upon landing). This injury process is depicted in [figure 1.1a](#), in which a margin of safety is observed in the first cycle of subfailure load. In the second loading cycle, the applied load increases in magnitude, simultaneously decreasing the margin of safety to zero, at which point an injury occurs. Although this description of low back injury is common, particularly among medical practitioners who are required to identify an injury-causing event when completing forms for workers' compensation reports, my experience suggests that relatively few low back injuries occur in this manner.

Figure 1.1 (a) A margin of safety is observed in the first cycle of subfailure load. In the second loading cycle, the applied load increases in magnitude, simultaneously decreasing the margin of safety to zero, at which point an injury occurs. (b) The Canadian snowmobile driver (the author in this case from many years ago, who should have known better) is about to experience an axial compressive impact load to a fully flexed spine. A one-time application of load can reduce the margin of safety to zero because the applied load exceeds the strength or failure tolerance of the supporting tissues.



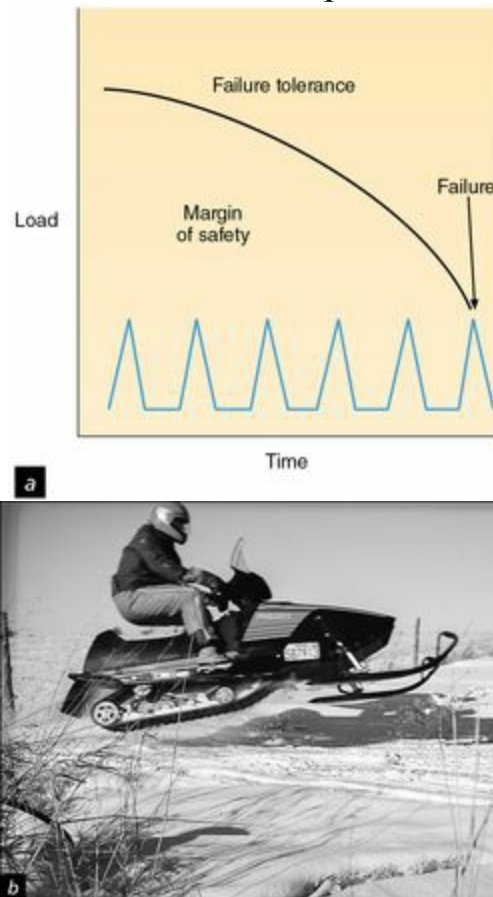
(a) Reprinted from Journal of Biomechanics, 30 (5), S.M. McGill, "Invited paper: Biomechanics of low back injury: Implications on current practice and the clinic," 456-475, 1997, with permission from Elsevier Science.

(b) Stuart McGill

More commonly, injury during occupational and athletic endeavors involves cumulative trauma from repetitive subfailure magnitude loads. In such cases, injury is the result of accumulated trauma produced by either the repeated application of relatively low load or the application of a sustained load for a long duration (as in a sitting task). A person lifting boxes onto a

pallet who is repeatedly loading the tissues of the low back (several tissues could be at risk) to a subfailure level (see [figure 1.2](#), a and b) experiences a slow degradation of failure tolerance (e.g., vertebrae; Adams and Hutton, 1985; Brinckmann, Biggemann, and Hilweg, 1989; Balkovec and McGill, 2012). As tissues fatigue with each cycle of load and correspondingly the failure tolerance lowers, the margin of safety eventually approaches zero, at which point this person will experience low back injury. Obviously, the accumulation of trauma is more rapid with higher loads (Callaghan, et al., 2012).

Figure 1.2 (a) Repeated subfailure loads lead to tissue fatigue, reducing the failure tolerance, leading to (b) failure on the Nth repetition of load, or box lift in this example.



(a) Reprinted from Journal of Biomechanics, 30 (5), S.M. McGill, "Invited paper: Biomechanics of low back injury: Implications on current practice and the clinic," 456-475, 1997, with permission from Elsevier Science.

(b) Stuart McGill

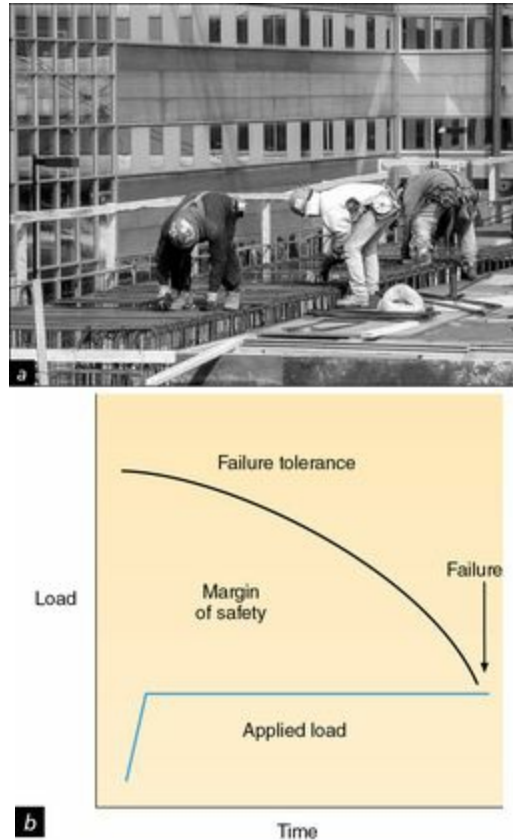
Yet another way to produce injury with a subfailure load is to sustain stresses constantly over a period of time. The rodmen shown in [figure 1.3a](#), with their spines fully flexed for a prolonged period of time, are loading the posterior passive tissues and initiating time-dependent changes in disc mechanics ([figure 1.3b](#)). Under sustained loads these viscoelastic tissues slowly deform and creep. The sustained load and resultant creep cause a progressive reduction in the tissue strength. Correspondingly, the margin of safety also declines until injury occurs at a specific percentage of tissue strain (i.e., at the breaking strain of that particular tissue). Note that these workers are not lifting a heavy load; simply staying in this posture long enough will

eventually result in injurious damage. The injury may involve a single tissue, or a complex picture may emerge in which several tissues become involved. For example, the prolonged stooped posture imposes loads on the posterior ligaments of the spine and posterior fibers of the intervertebral disc. The associated creep deformation that ultimately produces microfailure (e.g., Adams, Hutton, and Stott, 1980; McGill and Brown, 1992) may initiate another chain of events. Stretched ligaments increase joint laxity, which can lead to hyperflexion injury (to the disc) and to the following sequence of events:

1. Local instability
2. Injury of unisegmental structures
3. Ever-increasing shearing and bending loads on the neural arch

This laxity remains for a substantial period after the prolonged stoop (Solomonow, 2012).

Figure 1.3 (a) These rodmen with fully flexed lumbar spines are loading posterior passive tissues for a long duration, (b) reducing the failure tolerance leading to failure at the Nth% of tissue strain.



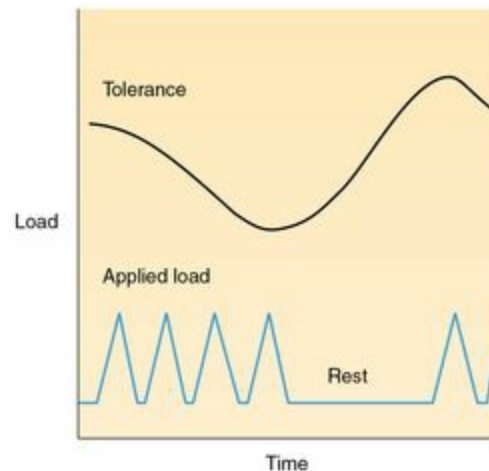
(a) Stuart McGill

Understanding the process of tissue damage in this way clarifies why simple injury prevention approaches often fail. More effective injury intervention strategies recognize and address the complexities of tissue overload.

The avoidance of loading altogether is undesirable. The objective of injury prevention strategies is to ensure that tissue adaptation stimulated from exposure to load keeps pace with, and ideally exceeds, the accumulated tissue damage. Thus, exposure to load is necessary, but in the process of accumulation of microtrauma, the applied loads must be removed (with rest) to allow the healing and adaptation process to gradually increase the failure tolerance to a higher level. We have already seen how tissue loading and injury risk form an optimal U-shaped relationship of not too much and not too little load. Determining the optimal load for health encompasses both the

art and science of medicine and tissue biomechanics. [Figure 1.4](#) presents a final load–time history to demonstrate the links among loading, rest, and adaptive tissue tolerance.

Figure 1.4 Loading is necessary for optimal tissue health. When loading and the subsequent degradation of tolerance are followed by a period of rest, an adaptive tissue response increases tolerance. Tissue “training” results from the optimal blend of art and science in medicine and tissue biomechanics.



In summary, the injury process may be associated with either very high loads or relatively low loads that are repeated or sustained. This either/or causation necessitates a rigorous examination of the injury and tissue loading history for substantial periods of time prior to the culminating injury event. It is important to recognize that simply focusing on a single variable such as one-time load magnitude may not result in a successful index of risk of injury, particularly across a wide variety of activities.

Excellent Clinicians and Excellent Practice

When I am asked to refer a patient to a clinician in a distant country, I try to think of a great one. They are rare indeed. Great clinicians perform assessments to uncover the cause of pain. Then they work to eliminate the cause first, followed by an intervention to reduce the risk of recurrence and enhance the robustness of the patient. This must include a component of moving well. Every system in the body requires movement for optimal function and performance. Few clinicians know how to select movement exercises, tune the dosage, and guide the progression.

Clinicians often dismiss the need for a thorough investigation of a patient's pain by stating that finding the pathological source is not possible. I believe that such clinicians are simply unknowledgeable and unskilled. Understanding injury pathways is nonnegotiable because this is the cornerstone of excellent practice. We know what causes disc herniations, Schmorl's nodes, spondylolisthesis, osteoporotic fractures, torn spine ligaments, facet arthritis, and stenosis, to name a few. Finding these causes is not always simple. Just as a homicide detective must amass enough circumstantial evidence to become convincing, the clinician tests diagnostic hypotheses with more follow-up tests until the evidence is overwhelming. The great clinicians work at the acceptable level of uncertainty. There is no such thing as nonspecific pain to the excellent clinician.

Assessment involves both art and science. Both are needed to interpret the interplay among neural, mechanical, psychological, and physiological variables. Each patient displays signs that must be interpreted. Some are false signs and need to be assessed with sham tests. Great clinicians have eclectic training experience and many tools in their toolboxes. Is specific exercise needed or manual tissue work, or perhaps both? Are the pain and symptoms from a single source or multiple sources? Are the radiating symptoms neural in origin (and if so, what is causing the nerve tension or compression), or are they ischemic? Is the pain organic to a tissue or a neural representation of pain similar to phantom limb pain (which I am convinced explains some of the myalgic syndromes)? These are just a few situations that are not dismissed by the excellent clinician. Worse yet are clinicians who fail to address the cause of the pain and then default to blame the patient for having a psychosocial disorder.

I close with one further generalization. A clinician who provides only passive treatments to a patient has little evidence documenting that the patient will have fewer symptoms in the future. For example, the clinician may perform an injection into the back (no supporting evidence for reducing back pain; Staal et al., 2009), manipulate some joints in the spine, or stretch the patient on a machine. Passive treatments may be helpful at the beginning of a larger treatment plan for certain people. In this case, the patient is progressed quickly into an active approach. The evidence does support the importance of the patients' being fully invested in understanding the cause of pain and practicing strategies throughout the day to avoid these pain triggers. Then they must commit to exercise approaches that suit their conditions.

Unique Scientific Foundation of This Book

This book contains many nontraditional viewpoints on how the spine functions and becomes damaged. Most of these perspectives have emerged from a unique biomechanically based methodological approach that we have been developing for over 30 years at the laboratory and clinic at the University of Waterloo in Canada. This chapter will familiarize you with the unique general approach to obtaining much of the data in this text and will help you understand both the limitations of and the unique insights provided by this approach.

As spine biomechanists, our methods of inquiry are similar to those used by mechanical or civil engineers. For example, a civil engineer charged with the task of building a bridge needs three types of information:

- Traffic to be accommodated, or the design load
- Structure to be used (e.g., space truss or Roman arch), because each architecture possesses specific mechanical traits and features
- Characteristics of the proposed materials that will affect strength, endurance, stability, resistance to structural fatigue, and so on

Our approach to investigating spine function is similar to that of our engineering colleagues. We begin with the following relationship to predict the risk of tissue damage:

$$\text{Applied load} > \text{tissue strength} = \text{tissue failure (injury)}$$

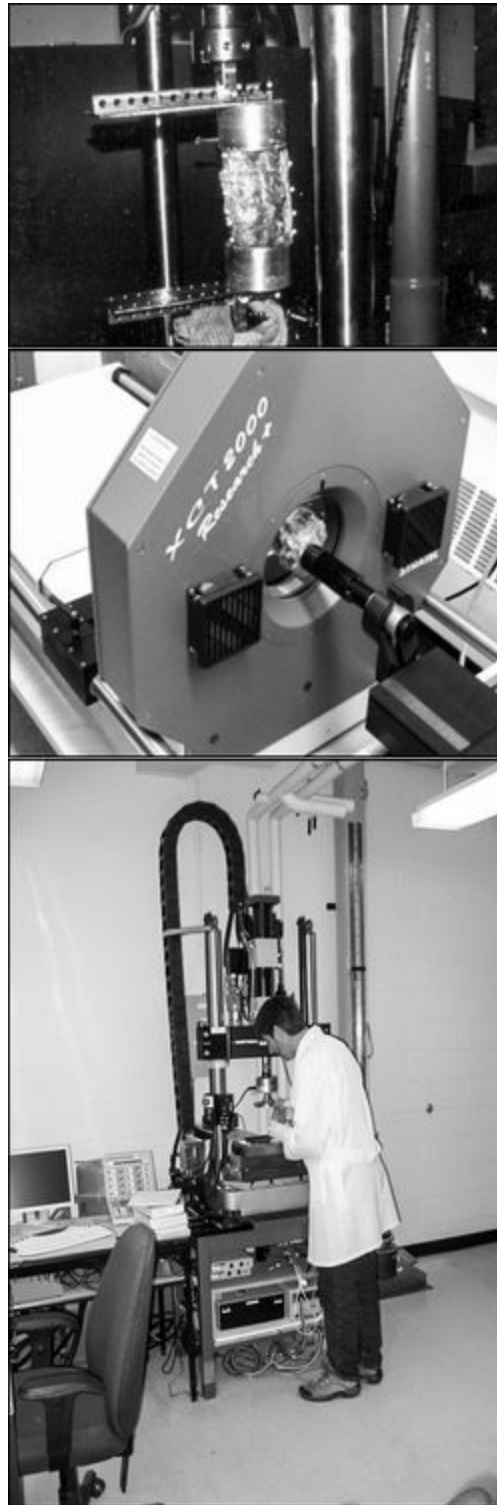
Recall from the tissue injury primer earlier in this chapter that tissue strength is reduced by repeated and prolonged loading but is increased with subsequent rest and adaptation. Analyzing tissue failure in this way requires two distinct methodological approaches. This is why we developed two quite distinct laboratories, which led to much of the progress documented in this book. (The “we” used in this chapter includes my research team of graduate students, visiting scholars, and technicians.) Our first lab is equipped for in vitro testing of spines, in which we purposefully try to create herniated discs, damaged end plates, and other tissue-specific injuries. The second lab is the in vivo lab, where living people (both those with and those without low back

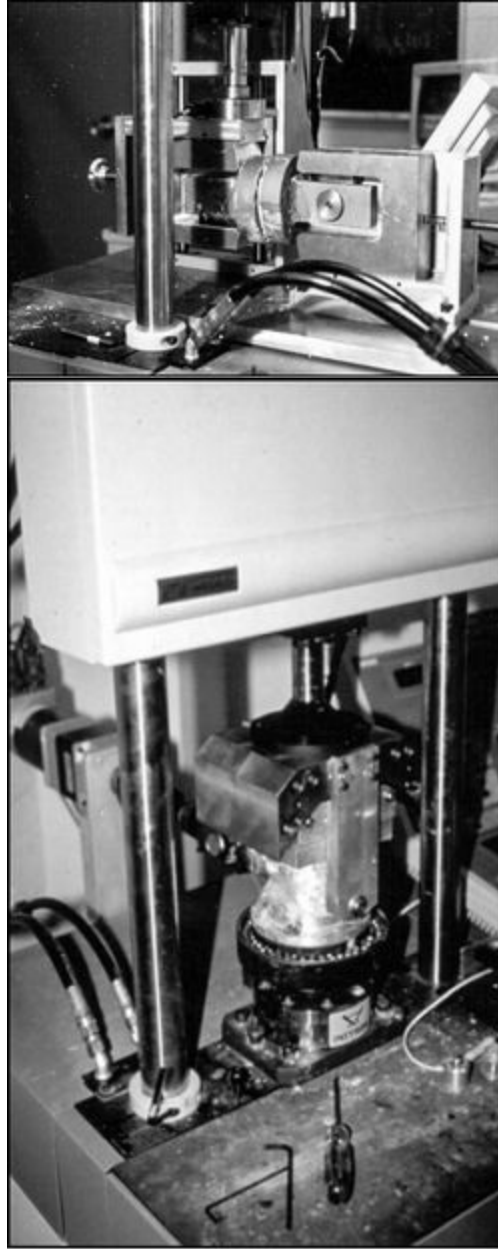
issues) are tested for their responses to stress and loading. Individual tissue loads are obtained from sophisticated modeling procedures. The final stage is to test the new findings in the clinic. Following is a brief explanation of the process.

In Vitro Lab

The in vitro lab is equipped with loading machines, an acceleration rack, tissue sectioning equipment, and an X-ray suite to document progressive tissue damage (see [figure 1.5](#)). For example, by performing discograms with radio-opaque contrast liquids, we can document the mechanics of progressive disc herniation. We investigate any other injury mechanisms in the same way—that is, by applying physiological loads and motion patterns and then documenting the damage with appropriate technology.

Figure 1.5 We employ many custom-made fixation jigs to mount and load spines. We then document the tissue damage with radiologic imaging and micro dissection.





Stuart McGill

Because many technique issues can affect the experimental results, the decision to use one over another is governed by the research question. For example, because a matched set of human spines to run a controlled failure test cannot be obtained, animal models must be used. Here, control is exercised over genetic homogeneity, diet, physical activity, and so forth, to contrast an experimental cohort with a matched set of control spines.

Of course, the results must be validated and interpreted to be relevant to humans. In addition, identifying the limitations for relevance in interpretation

is critical. Some hypotheses that demand the use of human material are compromised by the lack of available young, healthy, undegenerated specimens. Having healthy specimens is critical because biomechanics and injury mechanisms radically change with age. Other major methodological issues include the way biological tissues are loaded, perhaps at specific load rates or at specific rates of displacement. The researcher must decide which has the most relevance to the issue at hand. Devising these experiments is not a trivial task.

In Vivo Lab

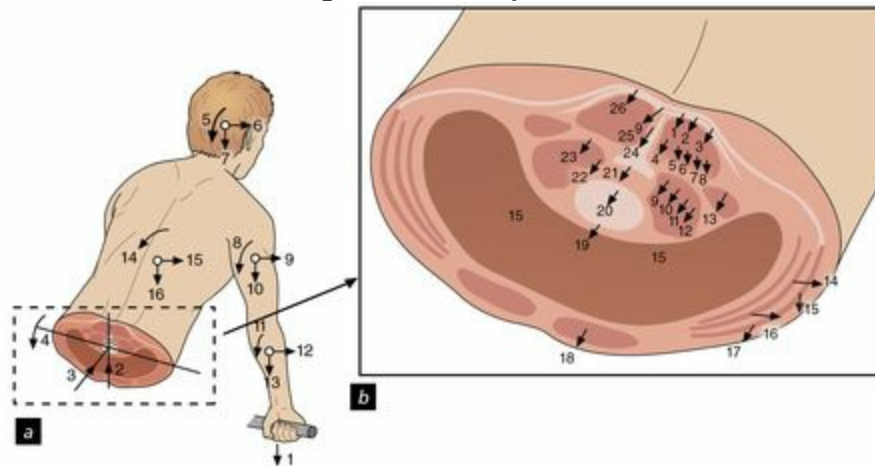
The in vivo lab is unique in its approach to documenting the loads on the many lumbar tissues in vivo. This knowledge lends powerful insight into spine mechanics, of both healthy and injured spines. Because transducers cannot be routinely implanted in the tissues to measure force, noninvasive methods are necessary. The intention of the basic approach is to create a virtual spine. This virtual model must accurately represent the anatomy that responds dynamically to the three-dimensional motion patterns of each test subject or patient and must mimic the muscle activation patterns chosen by the person. In so doing, it enables us to evaluate subjects' unique motor patterns and the consequences of their choices and skill.

How the Virtual Spine Works

Although two groups (the Marras group [e.g., Granata and Marras, 1993] and the McGill group) have devoted much effort to the development of biologically driven models, the McGill model is described here given its familiarity to the author. The model—a dynamic, three-dimensional, anatomically complex, biologically driven approach to predicting individual lumbar tissue loads—is composed of two parts: a linked-segment model and a highly detailed spine model that determines tissue loads and spine stability.

- The first part of the McGill model is a three-dimensional linked-segment representation of the body using a dynamic load in the hands as input. Two or more video cameras at 30 Hz record joint displacements to reconstruct the joints and body segments in three dimensions. Working through the arm and trunk linkage using linked-segment mechanics, reaction forces and moments are computed about a joint in the low back (usually L4-L5) (described in McGill and Norman, 1985) (see [figure 1.6a](#)). Using pelvic and spine markers, the three reaction moments are converted into moments about the three orthopedic axes of the low back (flexion–extension, lateral bend, and axial twist).

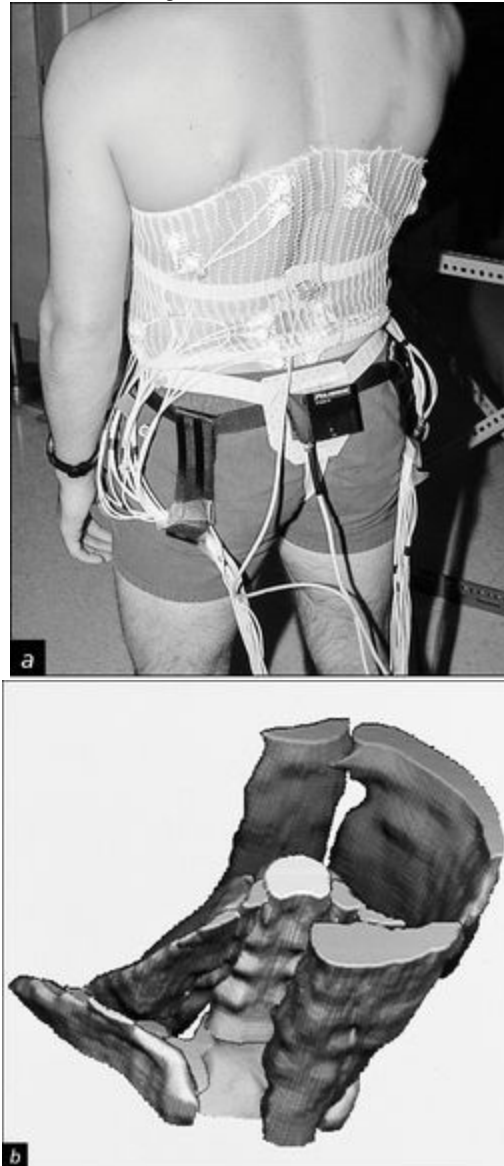
Figure 1.6 The tissue load prediction approach requires two models. (a) The first is a dynamic three-dimensional linked-segment model to obtain the three reaction moments about the low back. (b) The second model partitions the moments into tissue forces (muscle forces 1-18; ligaments 19-26; and moment contributions from deformed disc, gut, and skin in bending). These forces are applied to the various components, and the stresses are calculated and compared to the structure strengths to assess the risk of pain and injury. Finally, muscle stiffness is calculated, which allows the measurement of spine stability.



- The second part of the McGill model enables the partitioning of the reaction moments obtained from the linked-segment model into the substantial restorative moment components (supporting tissues) using an anatomically detailed three-dimensional representation of the skeleton, muscles, ligaments, nonlinear elastic intervertebral discs, and so on (see [figure 1.6b](#)). This part of the model was first described by McGill and Norman (1986); full three-dimensional methods were described by McGill (1992). Several anatomical additions and updates were provided by Cholewicki and McGill (1996); Vera Garcia, Moreside, and McGill (2009); Brown and McGill (2009); Grenier and McGill (2008); and Brown and McGill (2008a, 2008b). In total, over 120 back and torso muscles are represented. Very briefly, first the passive tissue forces are predicted by assuming stress–strain, or load deformation, relationships for the individual passive tissues. Deformations are modeled from the three-dimensional lumbar kinematics measured from the subject, which

drive the vertebral kinematics of the model. Passive tissue stresses are calibrated for the differences in flexibility of each subject by normalizing the stress–strain curves to the passive range of motion of the subject. Electromagnetic instrumentation, which monitors the relative lumbar angles in three dimensions, detects the isolated lumbar motion. The remaining moment is then partitioned among the many fascicles of muscle based on their activation profiles (measured from electromyography [EMG]) and their physiological cross-sectional area. The moment is then modulated with known relationships for instantaneous muscle length of either shortening or lengthening velocity. Sutarno and McGill (1995) described some improvements of the force–velocity relationship. In this way, the modeled spine moves according to the movements of the subject’s spine, and the virtual muscles are activated according to the activation measured directly from the subject (see figures 1.7, a and b, and 1.8).

Figure 1.7 (a) A subject monitored with EMG electrodes and electromagnetic instrumentation to directly measure three-dimensional lumbar kinematics and muscle activity. (b) The modeled spine (partially reconstructed for illustration purposes, although for the purposes of analysis it remains in mathematical form) moves in accordance with the subject's spine. The virtual muscles are activated by the EMG signals recorded from the subject's muscles.



Photos from Stuart McGill

Figure 1.8 (a) In this historical photo, an instrumented subject simulates the complex three-dimensional task of tossing an object. The instrumentation includes three-dimensional video to capture body segment kinematics, a recording of the three-dimensional force vector applied to the hand, a 3-Space electromagnetic device to record isolated three-dimensional lumbar motion and assist in partitioning the passive tissue forces, and 16 channels of EMG electrodes to capture muscle activation patterns. (b) A more modern data collection is shown on the right, where force plates register the forces under the feet to begin mapping the forces up to the spine. Infrared cameras (12 of them around the lab room) reconstruct the body in 3-D, and EMG electrodes record muscle activity.



Photos from Stuart McGill

Using biological signals in this fashion to solve the indeterminacy of multiple load-bearing tissues facilitates the assessment of the many ways we choose to support loads. Such an assessment is necessary for evaluating injury mechanisms and formulating injury-avoidance initiatives. From a clinical perspective, this ability to mimic individual spine motions and muscle activation patterns enables us to evaluate the consequences of a chosen motor control strategy. For example, we can see that some people can stabilize their backs and spare their lumbar tissues from overload when performing specific tasks. Conversely, we can evaluate the consequences of poorly chosen motor strategies. In this way we can identify people with perturbed motor patterns and devise specific therapies to regroove healthy motor patterns that ensure sufficient spine stability and spare their tissues from damaging load.

Our challenge has been to ensure sufficient biological fidelity so that estimations of tissue forces are valid and robust over a wide variety of activities. The three-dimensional anatomy is represented in computer memory (muscle areas are provided in appendix A.1). On occasion, if the expense is warranted, we create a virtual spine from a three-dimensional reconstruction of serial magnetic resonance imaging (MRI) slices from the hip trochanter to T4 (see [figure 1.7b](#)). This component of the modeling process is well documented for the interested reader in McGill and Norman (1986) and McGill (1992). A list of the large number of associated research papers pertaining to the many detailed aspects of the process is provided in the references and additional readings section at the end of this book. See [figures 1.9](#) and [1.10](#) for a flowchart and example of the modeling process.

Figure 1.9 The model input and output are illustrated in this flowchart up to the point of the calculation of moments and tissue loads. Spine stability is calculated with an additional module.

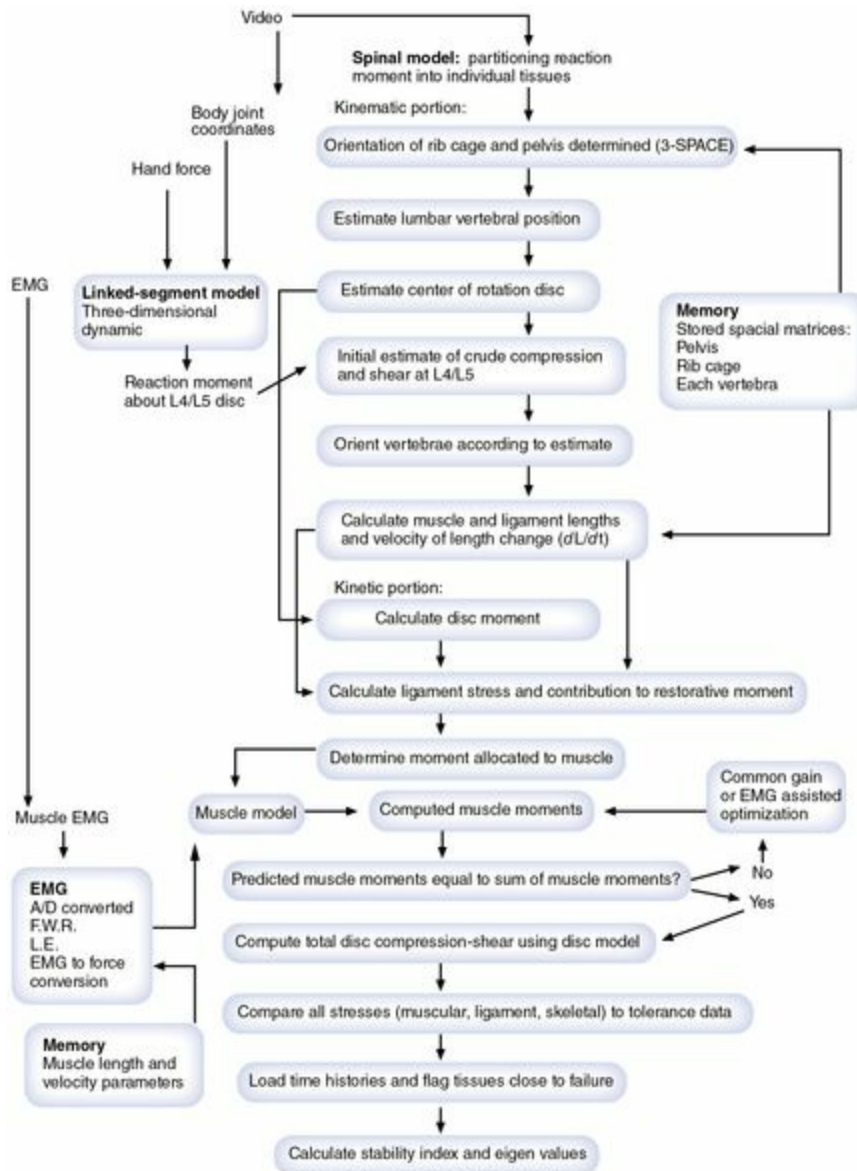
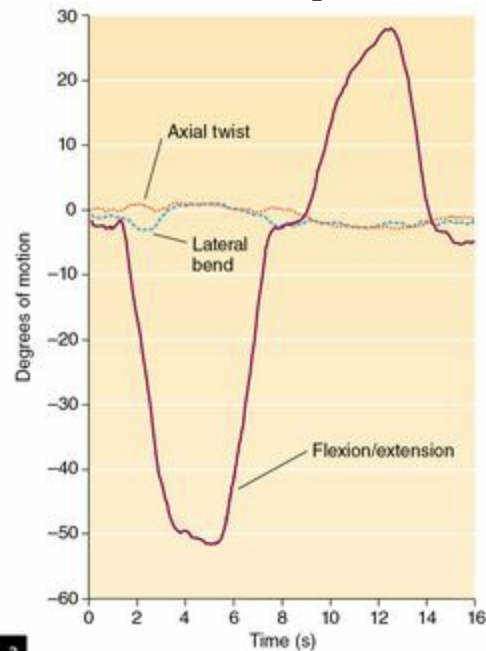
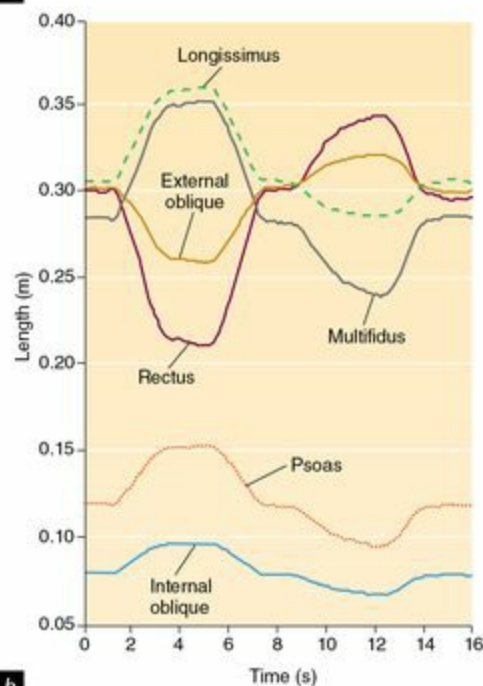


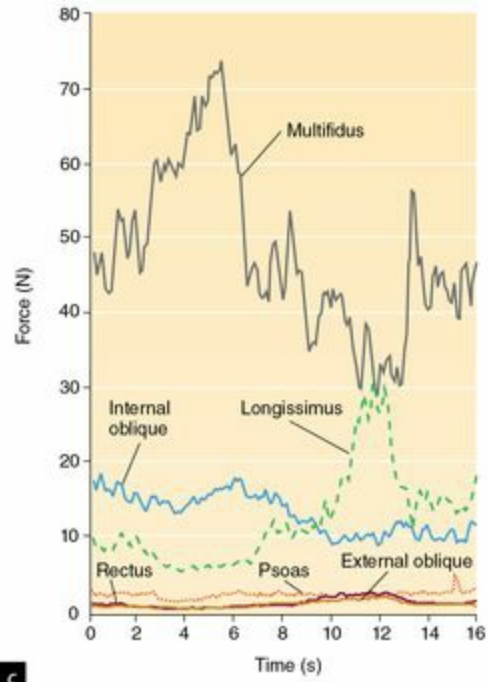
Figure 1.10 Stages of model output in this example of a subject flexing, picking up a weight, and extending: (a) lumbar motion about the three axes—flexion–extension, lateral bend, and axial twist; (b) lengths of a few selected muscles throughout the motion; (c) some muscle forces; (d) L4-L5 joint forces of compression and shear; (e) a stability index, in which larger positive numbers indicate higher stability, and a zero or negative number suggests that unstable behavior is possible.



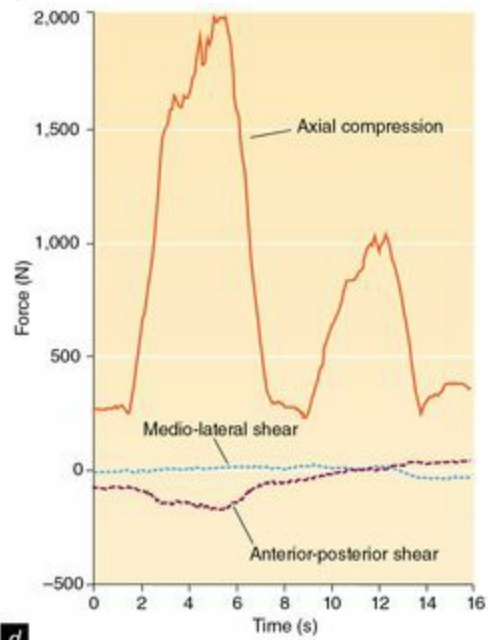
a



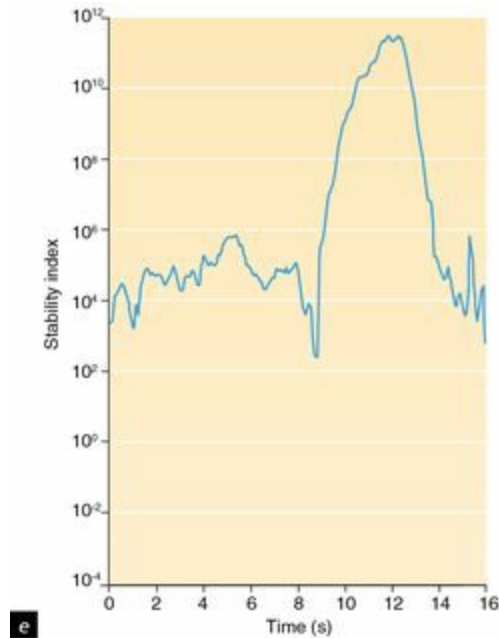
b



c



d



Development of the Virtual Spine

The development of the virtual spine approach has been an evolutionary process spanning 20 years. In that time we have had to confront issues of validity and how to handle deep muscles that are inaccessible with surface EMG electrodes. Briefly, although we have tried intramuscular electrodes in highly selected conditions, this is a limiting invasive procedure. Generally, we estimate the deep muscle activation amplitudes from movement synergists (McGill, Juker, and Kropf, 1996). This method is limited, however, because it requires prior knowledge of muscle patterns for a given moment combination in a specific task. We try to incorporate the highest level of content validity by using detailed representations of the anatomy and physiological cross-sectional areas, recording stress–strain relationships, and incorporating known modulators of muscle force such as length and velocity. One of our validation exercises is to compare the three measured reaction moments with the sum of individual tissue moments predicted by the virtual spine. A close match suggests that we have succeeded in accurately representing force.

Over the past 15 years, this approach for predicting individual lumbar tissue loads has evolved to enable us to document spine stability. In this way,

we can evaluate a person's motor patterns and identify strategies that ensure safety and those that could result in injury. This is not a trivial task. It requires converting tissue forces to stiffness and using convergence algorithms to separate the forces needed to create the torques that sustain postures and movements from the additional forces needed to ensure stability. Potential energy-based analyses are then employed to identify the stability index in each degree of freedom of the joint, which in turn reveals the joint's ability to survive a given loading scenario. This level of modeling represents the highest level of sophistication currently available. It has enabled us to challenge concepts pertaining to spine stability—more on this in chapters 5 and 10. The latest initiative, which is still in progress, is to complete the model to include legs via the hip joints and a cervical spine. This will extend our capability to assess the interplay with hip and neck function and provide a better understanding of disorders.

A Final Note

The controversies introduced in this chapter illustrate the need for the evidence presented in the rest of the text and the relevance of the discussions that follow. The scientific approaches used to build the basic scientific foundation are unique, which assist for interpreting the patient trials that guide how the clinical approaches are designed and prescribed. Resist the urge to assume that conventional wisdom is correct: First consider the evidence, and then form your own opinions.